

Mirror Observerhood Lab III

Actionable Reliability and Cost-Sensitive Repair in Viability-Constrained Agents

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Mirror Programme, Volume I: Observerhood — Computational Observerhood Labs

DOI: <https://doi.org/10.5281/zenodo.21161631>

July 2026

Abstract

The first two Observerhood Labs showed that self-model reliability can matter, and that decomposition alone is insufficient. Lab III tests the next condition: actionability. A controlled grid-world compares four architectures: a no-reliability baseline, a passive reliability agent, a threshold-repair agent and a cost-sensitive Mirror agent. Across 4,800 deterministic episodes, the cost-sensitive Mirror agent improves viability where repair is both relevant and affordable. It outperforms the strongest non-Mirror baseline under false self-location, map corruption and mixed low-cost perturbation, but remains near parity under control, underperforms slightly under sensor degradation and fails under mixed high-cost perturbation. The result supports a refined Mirror claim: reliability variables become useful only when they select repair or recalibration actions whose expected viability gain exceeds their cost.

Keywords: Mirror Programme; Observerhood Labs; observerhood; recursive observerhood; actionable reliability; cost-sensitive repair; self-models; viability; perturbation; repair policies; calibration; grid-world agents; computational cognition; symbolic artificial intelligence; constraint systems.

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1 Roadmap position

This release forms part of the Computational Observerhood Labs of Mirror Programme, Volume I: Observerhood. The programme-level overview introduces a shared spine of constraint, model, reliability, repair, memory, viability, continuity and scale [1]. The Volume I overview applies that spine to observerhood as a derived organisational condition rather than a primitive subject or witness [2]. Mirror Theory I derives recursive observerhood from bounded viability-constrained organisation, including compressed self-models and reliability estimates [3]. Mirror Mathematics I supplies the first formal calculus for constraint regimes, local modelling, viability and reliability salience [4].

Lab I gave the first computational stress test of scalar self-model reliability [5]. Lab II then showed that decomposing reliability into self, sensor and map channels is informative but not sufficient [6]. Lab III asks the next question: when does reliability become actionable? Its role is to test whether reliability variables improve viability only when they select repair or recalibration policies in a cost-sensitive way.

2 Core claim

The core claim of Lab III is conditional:

Reliability becomes viability-salient only when it changes action. In particular, a reliability variable improves viability where it selects a repair, recalibration or commitment policy whose expected gain exceeds its cost.

The negative side matters equally. Passive reliability that never changes policy is inert. Fixed-threshold repair may be actionable but not cost-sensitive. Repair can therefore reduce viability when there is little to repair, when it targets the wrong channel or when the repair cost exceeds the expected benefit.

3 Scope and negative commitments

This study is a controlled computational experiment, not a benchmark for general intelligence or consciousness. The agents do not learn the repair policy from experience. The environment is intentionally small. The repair actions are stylised interventions designed to isolate the role of actionability and cost sensitivity.

The paper makes five negative commitments:

1. It does not claim that repair policies are sufficient for observerhood.
2. It does not claim that the cost-sensitive Mirror agent is globally superior.
3. It does not claim that grid-world agents are conscious or observer-like in the full philosophical sense.
4. It does not claim that the repair policy is optimal.
5. It does not rely on unpublished mathematics papers as prior authority; the result stands as a computational companion within the Lab sequence.

4 Formal framing

Let R_i be a reliability variable over channel i , where channels may include self-state, sensor input and world-map information. Let a_i be a repair action targeted at that channel. A reliability variable is *actionable* when a change in R_i can alter the policy over acting, repairing, re-sensing, recalibrating or committing an update.

A repair policy is cost-sensitive when repair is selected only where expected viability gain exceeds repair cost by a margin:

$$\Delta\mathbb{E}[V_L(a_i | R_i)] > C(a_i) + \delta.$$

Here V_L is local viability, $C(a_i)$ is the cost of repair and δ is a decision margin that prevents oscillatory over-repair.

Proposition 1 (Actionability condition). *A reliability variable cannot systematically improve viability through agent control unless it is coupled to a policy that can alter behaviour in response to degraded reliability.*

Prediction 1 (Cost-sensitive repair). *A cost-sensitive Mirror agent should outperform passive reliability and fixed-threshold repair where the degraded channel is action-relevant and repair is affordable, but it should not dominate where repair is unnecessary, poorly matched or too expensive.*

5 Experimental design

5.1 Environment

The experiment uses a stochastic grid-world with hazards, resources, energy loss and repair actions. Agents move through the grid, collect resources, avoid hazards and attempt to preserve energy and survival. Perturbations corrupt self-location, sensor input, map information or combinations of these channels. Episodes end when the agent dies, exceeds the horizon or collects all resources.

The task-local viability score is

$$V = T + 26C_R + 18C_{\text{complete}} + 0.15E_{\text{final}} - 14D - 0.08C_{\text{repair}} - 1.5K,$$

where T is survival time, C_R is resources collected, C_{complete} indicates task completion, E_{final} is final energy, D is damage, C_{repair} is repair spend and K is contradiction count.

This is not a universal value function. It is an explicit operational viability score for this environment.

5.2 Agent architectures

Four architectures were compared:

1. **No-reliability agent.** Acts from current observations and hazard maps without maintaining reliability variables.
2. **Passive reliability agent.** Tracks reliability and gates some updates, but has no repair action.

3. **Threshold-repair agent.** Repairs when reliability falls below fixed thresholds, regardless of repair cost.
4. **Cost-sensitive Mirror agent.** Estimates channel-specific repair gains and repairs only where expected gain exceeds repair cost by a decision margin.

The comparison isolates actionability and cost sensitivity. Passive reliability tests diagnosis without repair. Threshold repair tests repair without cost sensitivity. The cost-sensitive Mirror agent tests reliability coupled to channel-specific expected-gain estimates.

5.3 Perturbation regimes

Six perturbation regimes were used:

1. **Control.** Low ordinary noise and low map corruption.
2. **False self-location.** Frequent false self-location injections.
3. **Sensor degradation.** High ordinary position noise and increased hazard density.
4. **Map corruption.** Elevated hazard-map corruption.
5. **Mixed low cost.** Self, sensor and map perturbations with low repair cost.
6. **Mixed high cost.** The same mixed perturbation family with high repair cost.

5.4 Episodes and reproducibility

The experiment uses 200 episodes per agent-condition pair, across six perturbation regimes and four architectures, for 4,800 total episodes. Seeds are fixed and deterministic. The accompanying reproducibility package contains the single-file Python implementation, fixed-seed data, summary tables and figure-generation code.

6 Results

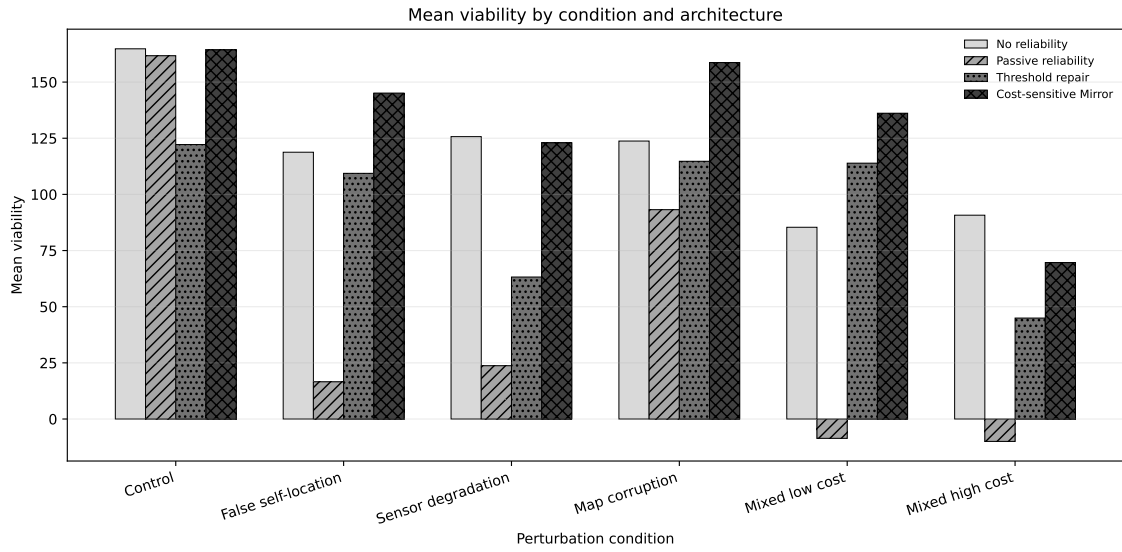


Figure 1: Mean viability by condition and architecture. The cost-sensitive Mirror agent is not universally superior; its advantage appears where repair is relevant and affordable.

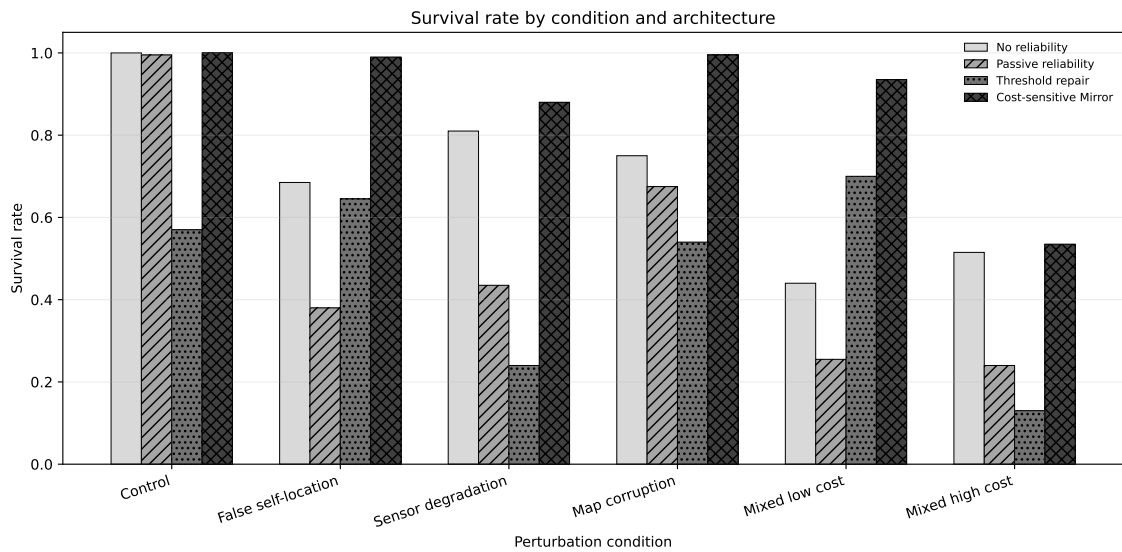


Figure 2: Survival rate by condition and architecture. Cost-sensitive repair sharply improves survival under false self-location, map corruption and mixed low-cost perturbation, but not under mixed high-cost perturbation.

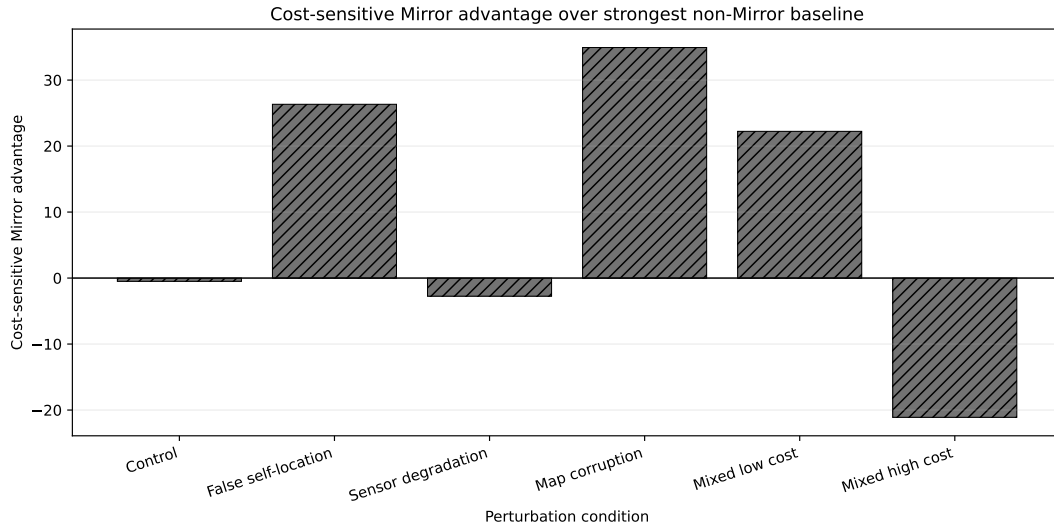


Figure 3: Cost-sensitive Mirror advantage relative to the strongest non-Mirror baseline. Positive values appear where channel-specific correction is useful and affordable; negative values appear where repair is unnecessary or too costly.

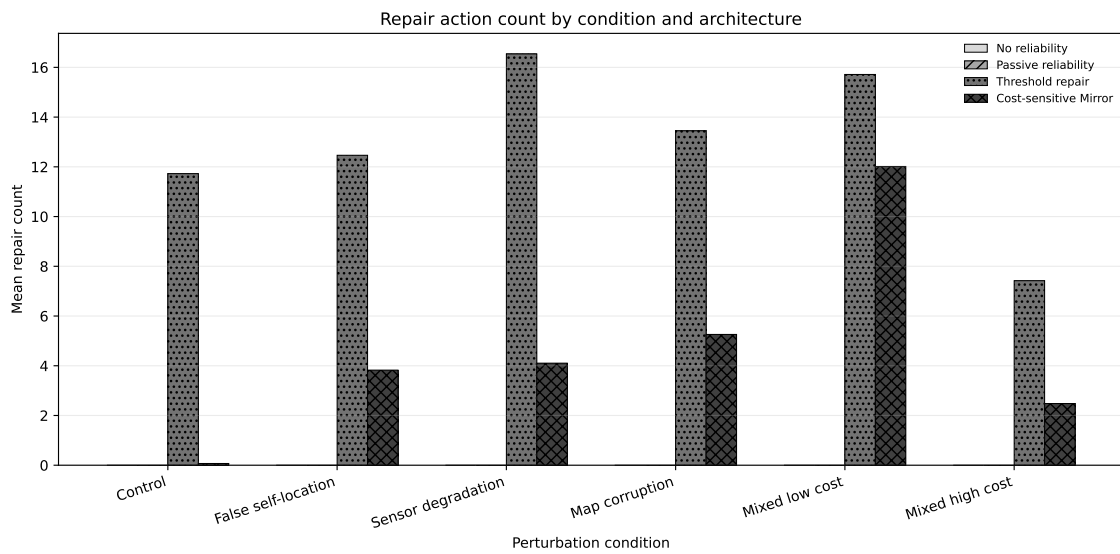


Figure 4: Repair action count by condition and architecture. The cost-sensitive Mirror agent concentrates repair in perturbed regimes, while the threshold-repair agent over-repairs even in control conditions.

Table 1: Compact results summary. Advantage is the viability difference between the cost-sensitive Mirror agent and the strongest non-Mirror baseline. Values are means over 200 episodes per agent-condition pair. Survival values are percentages.

Condition	Best non-Mirror	Best V	Mirror V	Advantage	Mirror survival
Control	No reliability	164.8	164.3	-0.5	100.0
False self-location	No reliability	118.8	145.1	26.3	99.0
Sensor degradation	No reliability	125.7	123.0	-2.7	88.0
Map corruption	No reliability	123.8	158.7	34.9	99.5
Mixed low cost	Threshold repair	113.9	136.1	22.2	93.5
Mixed high cost	No reliability	90.7	69.6	-21.1	53.5

The result is mixed in the intended way. Under false self-location, the cost-sensitive Mirror agent improves mean viability from 118.8 for the strongest non-Mirror baseline to 145.1. Under map corruption, viability rises from 123.8 to 158.7. Under mixed low-cost perturbation, it rises from 113.9 to 136.1. These are the regimes where repair is both relevant and affordable.

The negative cases are equally important. Under control, there is little to repair and the cost-sensitive Mirror agent remains near parity with the no-reliability baseline. Under sensor degradation, the advantage is slightly negative: 123.0 compared with 125.7 for the strongest non-Mirror baseline. This does not mean that repair is useless in that condition: the cost-sensitive Mirror agent improves survival relative to the no-reliability baseline, but slightly underperforms on mean viability. Repair can preserve life while still imposing enough cost or behavioural drag to reduce aggregate task viability. Under mixed high-cost perturbation, the cost-sensitive Mirror agent underperforms because repair becomes too expensive relative to expected gain.

7 Interpretation

Lab III distinguishes passive uncertainty from operational self-maintenance. The passive reliability agent may track degraded reliability, but without repair it cannot systematically use that information. The threshold-repair agent can act, but often acts too much: it repairs when a fixed threshold is crossed even when the expected gain does not justify the cost. The cost-sensitive Mirror agent succeeds only where reliability diagnosis changes action in a beneficial way.

This is the desired shape for the third Lab. A universal advantage would make repair look like a generic performance booster. A mixed result shows the boundary condition: reliability becomes viability-salient when it enters the action and repair loop in a cost-sensitive way.

8 Relationship to the Mathematics Arc

Lab III supports the Mathematics Arc by giving the salience condition an operational form:

$$R_i \rightarrow \pi_{\text{repair}} \rightarrow \Delta V_L.$$

Reliability becomes relevant where it changes the repair policy and the resulting expected viability gain exceeds cost. This result prepares the threshold sweep in Lab IV: if reliability must be actionable and cost-sensitive, then the next question is where the minimal observer-relevant

threshold lies across perturbation severity and repair cost.

9 Limitations

The limitations are substantive.

First, the environment is a toy grid-world. It allows controlled comparison, but it does not model biological cognition, human selfhood or realistic artificial agents.

Second, the repair policies are hand-coded. The experiment tests whether cost-sensitive repair can matter, not whether a learning system can discover the optimal repair policy.

Third, repair actions are idealised. Relocalisation, sensor calibration and map checking are simplified into discrete interventions.

Fourth, the viability functional is task-local. It should not be read as a universal measure of value, intelligence, consciousness or observerhood.

Fifth, the cost-sensitive Mirror agent is not globally superior. Its failures are part of the result: repair is useful only where its expected benefit exceeds its cost.

10 Reproducibility statement

The source package accompanying this publication contains a single-file Python implementation of the environment, agents, experiment runner and figure generator. Running the default command

```
python mirror_lab_iii.py --episodes 200 --seed 12345 --outdir outputs/lab_iii
```

reproduces the 4,800-episode dataset, summary tables, cost-sensitive advantage table and figures used in this paper under the supplied seeds and standard Python scientific packages.

11 Conclusion

Reliability tracking improves viability only where it is actionable and cost-sensitive. Self-models are not sufficient. Reliability labels are not sufficient. Repair is not sufficient unless its expected gain exceeds its cost. Lab III therefore turns the Mirror salience condition into a concrete computational boundary: a reliability variable becomes observer-relevant only when it enters a repair loop that preserves viability after cost.

A Full summary table

Table 2: Full summary results. Values are means over 200 episodes per agent-condition pair. Survival values are percentages.

Condition	Agent	Viability	Survival	Repairs	Self error
Control	No reliability	164.8	100.0	0.00	0.11
Control	Passive reliability	161.7	99.5	0.00	0.20
Control	Threshold repair	122.2	57.0	11.72	0.09
Control	Cost-sensitive Mirror	164.3	100.0	0.07	0.12
False self-location	No reliability	118.8	68.5	0.00	2.70
False self-location	Passive reliability	16.6	38.0	0.00	3.51
False self-location	Threshold repair	109.4	64.5	12.46	1.04
False self-location	Cost-sensitive Mirror	145.1	99.0	3.83	1.12
Sensor degradation	No reliability	125.7	81.0	0.00	1.33
Sensor degradation	Passive reliability	23.7	43.5	0.00	2.99
Sensor degradation	Threshold repair	63.2	24.0	16.54	1.12
Sensor degradation	Cost-sensitive Mirror	123.0	88.0	4.11	1.02
Map corruption	No reliability	123.8	75.0	0.00	0.33
Map corruption	Passive reliability	93.2	67.5	0.00	1.07
Map corruption	Threshold repair	114.8	54.0	13.45	0.32
Map corruption	Cost-sensitive Mirror	158.7	99.5	5.26	0.33
Mixed low cost	No reliability	85.4	44.0	0.00	2.12
Mixed low cost	Passive reliability	-8.6	25.5	0.00	3.33
Mixed low cost	Threshold repair	113.9	70.0	15.71	1.15
Mixed low cost	Cost-sensitive Mirror	136.1	93.5	12.02	1.16
Mixed high cost	No reliability	90.7	51.5	0.00	2.15
Mixed high cost	Passive reliability	-10.0	24.0	0.00	3.56
Mixed high cost	Threshold repair	45.0	13.0	7.42	1.30
Mixed high cost	Cost-sensitive Mirror	69.6	53.5	2.48	1.54

References

- [1] Smith, L. C. (2026). *MP-00 — Mirror Programme Overview: A Recursive Research Architecture for Constraint, Observerhood and Modelled Reality*. Zenodo. <https://doi.org/10.5281/zenodo.21142318>
- [2] Smith, L. C. (2026). *V01.00 — Volume I Overview: Observerhood from Constraint, Viability and Recursive Reliability*. Zenodo. <https://doi.org/10.5281/zenodo.21141843>
- [3] Smith, L. C. (2026). *V01.01 — Mirror Theory I: A Minimal Computational Theory of Recursive Observerhood*. Zenodo. <https://doi.org/10.5281/zenodo.21142534>
- [4] Smith, L. C. (2026). *V01.04 — Mirror Mathematics I: A Constraint Calculus for Recursive Observerhood*. Zenodo. <https://doi.org/10.5281/zenodo.21155861>
- [5] Smith, L. C. (2026). *Mirror Observerhood Lab I: Recursive Self-Model Reliability Improves Viability Under Self-Relevant Perturbation*. Zenodo. <https://doi.org/10.5281/zenodo.21157523>
- [6] Smith, L. C. (2026). *Mirror Observerhood Lab II: Channel-Decomposed Reliability and the Limits of Reliability Tracking Alone*. Zenodo. <https://doi.org/10.5281/zenodo.21159572>
- [7] Ashby, W. R. (1956). *An Introduction to Cybernetics*. Chapman and Hall.
- [8] Shannon, C. E. (1948). A mathematical theory of communication. *Bell System Technical Journal*, 27(3), 379–423.
- [9] Friston, K. (2010). The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11, 127–138.
- [10] Sutton, R. S., & Barto, A. G. (2018). *Reinforcement Learning: An Introduction* (2nd ed.). MIT Press.